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The benefits of yoga as an intervention for depression: a meta-analysis

Abstract

Depression is a major societal issue that affects a large proportion of the population. Yoga has been introduced as a viable alternative approach to reduce symptoms. While prior studies have used meta-analysis to examine the benefits of yoga for depression, these studies incorporated a small number of studies and were limited in their ability to conduct a sub-group analysis. The purpose of this study was to quantify the effect size of yoga in the treatment of depression. Inclusion criteria included experimental, quasi-experimental, and feasibility studies that examined yoga with physical postures (asanas) as a treatment for depression. Nine moderator variables were examined including type of yoga, components of yoga (i.e., breathing, meditation, or relaxation), experience of yoga instructor, depression diagnosis, type of comparison group, research design, and length of treatment. Risk of bias was assessed using the Jadad instrument. Electronic databases were used to identify studies, and the search concluded on January 31st, 2022. A total of 152 studies (150 articles) met the inclusion criteria in this meta-analysis with 8210 participants assessed. The overall weighted effect size of yoga for treatment of depression was Hedges' $g = 0.55$ (95% CI 0.44–0.65), demonstrating a medium effect. Meta-regression and subgroup analyses found larger effect sizes as the amount of yoga practiced increased, when relaxation was practiced, and when participants had been clinically diagnosed with depression. Funnel plot and trim-and-fill procedure found little evidence of publication bias and the fail-safe number was 21783. These results demonstrate that yoga is an effective integrative approach in the treatment of depression. This study was limited by the methodological quality of the individual studies that make up the larger analysis, including participants who were not blind to the procedure and a failure to report withdrawals and dropouts.

Keywords: asanas, depression, complementary and alternative medicine, mind-body intervention, moderator analysis

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Abbreviations: RCT, randomized controlled trial; PRISMA preferred reporting items for systematic reviews and meta-analyses; IG, independent groups; PP, pretest/posttest; PPC, pretest/posttest with control; IRQ, interquartile range; CIs, confidence intervals; ES, effect size; SE, standard error

Introduction

Psychological disorders are a rampant global problem, with mental illness accounting for one-third of global disability.¹ In a national survey of U.S. adults, the lifetime prevalence of major depressive disorder was estimated at 20.6%.² During the COVID-19 pandemic, prevalence of depression rose from 3% to 25% globally.³ Drug therapy is a standard treatment approach which commonly serves to treat the symptoms of depression but does not fully treat or cure the disorder. Thus, individuals may benefit from the addition of a complementary or alternative therapy.⁴ A growing body of research has examined integrative and complementary medicine in the treatment of psychological disorders. For instance, yoga has been examined as an approach to enhance the efficacy of mental health interventions.

Yoga as an intervention

The first written records of yoga date back to 200 BC in Yogasutra of Patanjali.⁵ Traditionally, the goal of yoga was to unite mind, body, and spirit.⁶ In addition to the growing popularity of yoga, the scientific study of yoga practice has burgeoned over the past few decades. Yoga is sometimes used as a mind-body intervention for the treatment of depression. Prior systematic reviews have been conducted on the benefits of yoga for depression.^{7,8,9,10,11,12} Of these, three are meta-analyses.^{7,8,10} These meta-analyses, however, focused on randomized

controlled trials (RCTs) and excluded observational studies. While RCTs are considered the gold-standard in research methodology, there are several reasons why including non-RCT studies is advantageous. First, though RCTs theoretically reduce selection bias and confounding variables compared to non-RCTs, there is evidence that observational studies do not always overestimate treatment effects compared to RCTs.^{13,14} Second, since observational studies are not always more prone to bias, excluding non-RCTs lowers the precision of the estimated effect. Third, if study design is related to differences in effect sizes, then it is important to identify such discrepancies to make recommendations for future research. Finally, excluding studies limits the ability to examine potential moderator variables. Moderator analysis assesses the relationship between one or more covariates and an effect size.¹⁵ By identifying significant moderator variables, future research can be better informed and mental health treatments can be implemented more effectively. Therefore, non-RCTs were included with the goal of obtaining a more comprehensive estimate of the population effect size, to examine whether RCTs versus non-RCTs produce different effects, and to examine potential moderator variables.

Moderators

The following moderators were examined: type of yoga practiced, whether yoga included breathing, meditation, or relaxation, expertise of yoga instructor, depression diagnosis, type of comparison group, type of research design, and length of treatment. The nuances of yoga practice (i.e., type, frequency, and expertise of instructor) were selected for examination because a few studies have examined whether there are differential impacts when yoga is focused on

breathing (*pranayama*), relaxation, or meditation. Qi et al. (2020) found that college students benefited more from practicing yoga with a focus on breathing compared to yoga with a focus on meditation.¹⁶ In a review, researchers reported that improvements in blood pressure were only found when all three components of yoga were practiced (breathing, postures, and meditation).¹⁷ The research, however, has been insufficient in determining which aspect of yoga is most impactful for alleviating depression. The second moderator examined the experience of the yoga instructor; this moderator was included based on findings that yoga with a trained instructor was more effective compared to yoga practiced at home for treating trauma.¹⁸ Examining whether there are any moderating effects based on the type of yoga, expertise of yoga instructor, and the amount of yoga practiced has important implications for making appropriate recommendations.

Based on previous meta-analytic findings, the current study examined whether the type of depression diagnosis moderated the effect of yoga. A meta-analysis by Cramer et al.¹⁹ examined yoga as a treatment for anxiety. Findings indicated that when individuals were diagnosed by Diagnostic and Statistical Manual criteria there was no effect, but a small effect was found for diagnosis by other methods or when participants experienced symptoms without a diagnosis. Finally, as previously described, RCTs and non-RCTs were included to examine whether the research design moderated the impact of the yoga intervention. Prior meta-analyses that have examined the impact of yoga on depression have excluded non-RCTs; therefore, the current study has the benefit of examining whether the mean effect size differs based on study design.

Aim and objectives

The aim of this study was to determine the effectiveness of yoga in the treatment of depression. After a review of the literature, it was hypothesized that yoga is more effective under the following circumstances:

1. When more strenuous forms of yoga are practiced.
2. When yoga included breathwork (compared to no focused breathwork).
3. When yoga included meditation (compared to no focused meditation).
4. When yoga included relaxation (compared to no focused relaxation).
5. When yoga instructors who have more expertise are utilized.
6. When yoga is practiced for a longer duration.
7. When participants do not have a clinical diagnosis.
8. When compared to a no-treatment control group (e.g., wait list control group).
9. In non-RCT studies compared to RCT studies.

Material and methods

A meta-analysis was conducted which is often considered a superior approach compared to systematic or narrative reviews because the meta-analytic approach calculates a mean effect size¹⁵ and provides a more precise estimate of the magnitude of an effect compared to a single study. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines²⁰ were followed and this study was prospectively registered with the PROSPERO database (ID: CRD42020221400).

Literature search

The following electronic databases were searched to find published and nonpublished studies: Academic Search Premier, Google Scholar, PsycINFO, PubMed, and SportDiscus. The following key terms were used: *yoga*, *yoga therapy*, *yoga intervention*, *depression*. Studies were also located through a process of backward searching (i.e., using references of articles to track down older studies) and forward searching (i.e., entering the citation for an article into Google Scholar and selecting “as cited by” to locate newer studies). The search ended on January 31st of 2022.

Inclusion criteria

In guiding the selection of articles to be included in this meta-analysis, the following inclusion criteria were utilized:

1. The study had to examine yoga related to the treatment of depression.
2. The study had to implement a yoga program that involved physical postures (asanas).
3. The study had to be an experimental design, quasi-experimental design, or a feasibility study with an experimental condition and a control condition.
4. Manuscripts needed to have the necessary statistical values to compute effect sizes (e.g., sample sizes, sample means, and sample standard deviations).
5. The study had to be in English for the authors to read, code, and interpret the data.

An article inclusion flowchart is provided for a visual representation of the articles that were included in this meta-analysis (Figure 1). Using the above procedures, 150 papers were identified that met the inclusion criteria and 152 studies were included in the final analysis.

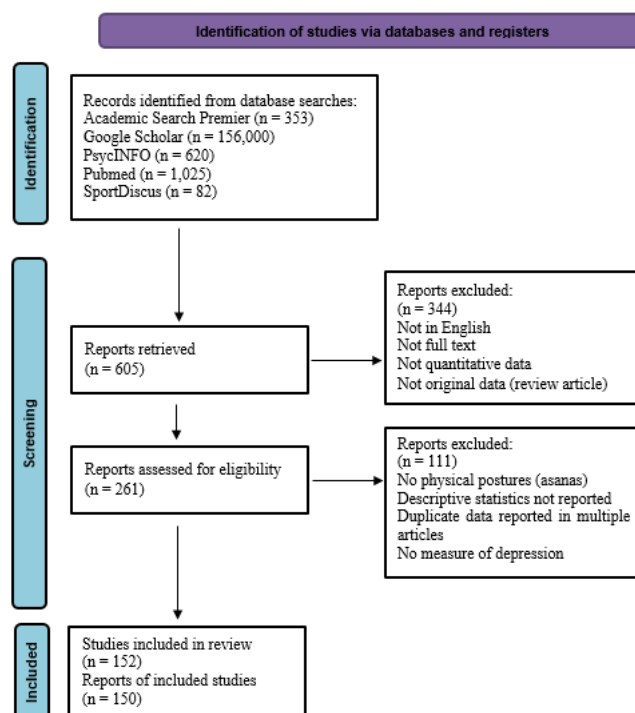


Figure 1 PRISMA Article Inclusion Flowchart.

Inter-rater reliability

To evaluate the accuracy of the coded variables, inter-rater reliability was analyzed. All variables were coded by the first author and by research assistants. After coder training was completed, the initial inter-rater agreement was 95%, with Cohen's kappa = .89. This initial reliability assessment using kappa is considered an "almost perfect" strength of agreement.²¹ All discrepancies were resolved between coders until 100% rater agreement was achieved.

Study coding procedures

Classifying the characteristics of the yoga program was the first step in the coding process (see Table 1). The first moderator variable coded was the type of yoga utilized. In the first round of coding, the exact type of yoga was recorded. Next, the most common types of yoga that appeared in the literature were identified. The five emergent categories were Swami Vivekananda Yoga, the Inner Resources Program, Kripalu, Iyengar, and Bikram yoga. A sixth "other" category was labeled for types of yoga that were not developed for the research study, or only appeared once in the literature. Further, each study was coded for whether the yoga practice focused on breathing, relaxation, or meditation.

The second moderator variable was the yoga instructor's level of expertise. Four categories were identified for this variable, including: 1) expertise not specified; 2) certification indicated, but hours of training not stated; 3) certified instructor with 200 hours of training; and 4) certified instructor with 500 hours of training or 5 years of teaching experience. The third moderator was length of treatment (how many hours yoga was practiced). This was calculated by multiplying session length with number of sessions.

Additionally, the type of depression diagnosis and the control group were coded. For depression diagnosis, studies were classified as including participants who were clinically depressed if this was used as the criterion for participation, versus studies where participants were not selected based on this criterion. For the type of control group, the possibilities included: 1) no-treatment (e.g., wait list or placebo control); 2) a comparison group (some alternative treatment was provided; e.g., psychoeducation, drug therapy, exercise); or 3) repeated measures (a pre-test post-test design was used in which participants served as their own controls). The final moderator variable coded for was whether the studies were RCTs versus non-RCTs.

Following procedures utilized by Wang et al.,¹⁷² methodological quality of each article was coded using the Jadad instrument (1996),¹⁷³ which evaluates randomization, blinding, and withdrawals/dropouts. Finally, the statistical values necessary to calculate effect sizes were coded. These values included the means, standard deviations, and sample sizes (*n*) of the experimental and control groups.

Statistical procedures

For each study, an effect size, which is a standardized difference in means, and its variance were calculated. The effect size and its variance are computed differently depending on the study design. The study designs fell into three categories:

1. Independent groups (IG). Each participant was included in one of two groups (experimental or control), but not both, and there were only post-intervention scores for the experimental group. Nine studies used this design.^{81,94,99,118,129–131,144,152}
2. Pretest/Posttest (PP). Participants formed a single group with repeated measures (pre- and post-intervention). Thirty-five

studies used this design.^{24,25,27,31,46,49,52,54,55,60,63,77, 78,84,87,93,95,101,109–112,119,122, 123,128,143,146,148,150,154,162–164,170}

3. Pretest/Posttest with Control (PPC). Each participant was included in one of two groups (experimental or control) with repeated measures (pre- and post-intervention) in both groups. One hundred six studies used this design.^{22,23,26,28–30,32–45,47,48,50,51,53,56–59,61,62,64–76, 79,80,82,83,85,86,88–92,96–98,100,102–108,113–117,120,121,124–127,132–142,145,147,149,151,153,155–161,165–169,171}

For IG studies, the effect size was Hedges *g*, which is the difference between the group means divided by the pooled standard deviation and corrected for bias.¹⁵ For PP studies, the standardized mean change using the pre-intervention standard deviation was used, correcting for bias.¹⁷⁴ For PPC studies, the effect size was the difference between the bias-corrected standardized mean change for the treatment and control groups.¹⁷⁵ In each case, a positive effect size corresponds to a favorable outcome.

For repeated measures studies (PP and PPC), the variance of the effect size decreases as the within-subject correlation between pre- and post-intervention depression scores, r_1 , increases.¹⁵ Furthermore, 14 studies reported multiple measures of depression for each participant.^{25,28,35,39,65,76,96,99,113,114,119,135,138,154} For each of these studies, the measure-specific effect sizes were averaged to obtain one combined effect size per study as recommended in the literature.¹⁵ The variance of the combined effect size for these studies increases as the within-subject correlation between different measures of depression, r_2 , increases.¹⁵ Because neither r_1 nor r_2 was reported in any of the studies, $r_1 = 0.5$ and $r_2 = 0.5$ were used, and a sensitivity analysis was conducted to assess the impact of these choices on the meta-analysis results.

Ray et al.¹³⁶ reported results separately for males and females, and Streeter et al.¹⁴⁸ reported results separately for two different yoga dose groups. Each of these articles were treated as two separate studies in the meta-analysis; one for males and the other for females (Ray et al.), and one for low-dose and the other for high-dose (Streeter et al.), as recommended by other authors.¹⁵ Chan et al.⁴⁰ and Pruthi et al.¹³⁰ reported the interquartile range (IQR) instead of the standard deviation. The standard deviations were approximated using the IQRs divided by 1.35 (based on the normal distribution ratio).

The overall summary effect was computed by fitting a random effects model (without moderator variables) to the primary studies' observed effects, and the null hypothesis of no overall effect was tested using the corresponding *z* test.¹⁵ Heterogeneity (variation in observed effect sizes beyond that due to sampling error) was quantified using the statistic T^2 . Heterogeneity was tested for using Cochran's *Q* test of whether T^2 differed significantly from zero.¹⁵

To assess the potential impact of heterogeneity in the meta-analysis, the I^2 statistic was used, which represents the percentage of the variation in observed effect sizes that is due to heterogeneity rather than sampling error.¹⁵ To determine which of the nine moderator variables contribute to heterogeneity, subgroup and meta-regression analyses were performed. These are analogous to ANOVA and regression on the observed effect sizes. The null hypothesis that the effect size is the same across different levels of the moderator variable (i.e., across different subgroups of studies or different numbers of total session hours) was tested using the Wald *Q* test (subgroup analyses) and the *z* test (meta-regression).¹⁵

Publication bias was examined using three methods. First, a funnel plot of the 152 studies was produced. A funnel plot examines whether

there is evidence that studies with small sample sizes were less likely to be published.¹⁷⁶ Second, a trim-and-fill procedure was conducted using the R_0 statistic.^{177,178} The trim-and-fill procedure provides an estimate of the number of studies missing from the funnel plot and provides a test of the null hypothesis that the number of missing studies is zero. Third, Rosenberg's fail-safe n , which provides an estimated number of studies with no effect that would reduce the mean effect size to non-significance, was calculated.¹⁷⁹ For all hypothesis tests, statistical significance was assessed using alpha level of .05. The meta-analysis was carried out in R using the "metafor" package.¹⁸⁰

Results

The observed effect size and characteristics of each study appear in Table 1, and the caterpillar plots in Figures 2 and 3 show the effect sizes with 95% confidence intervals (CIs). The overall mean effect size, based on the fitted random effects model, is $g = 0.55$ ($SE = 0.05$) with 95% CI 0.44–0.65. This is statistically significantly different from zero based on the z test ($z = 10.3$, $p < .001$).

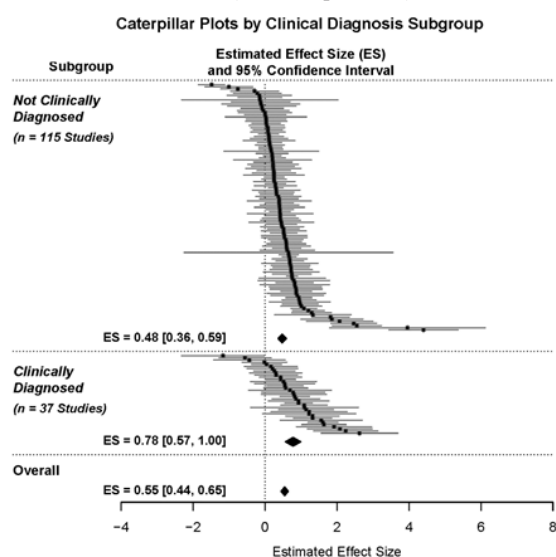


Figure 2 Caterpillar plots of the 152 primary studies split into clinical diagnosis subgroups.

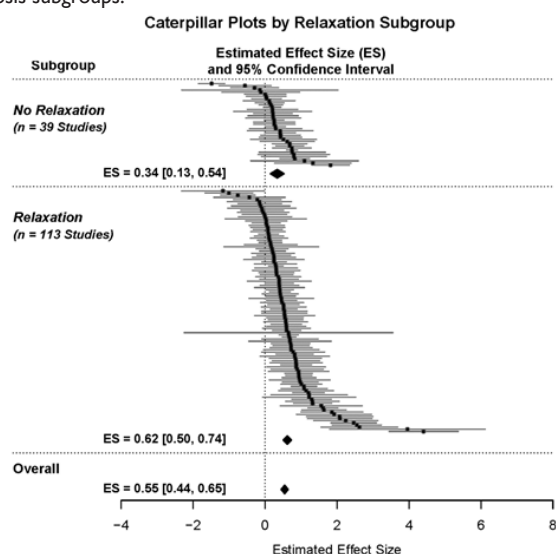


Figure 3 Caterpillar plots of the 152 primary studies split into relaxation subgroups.

Heterogeneity

The Q test for heterogeneity indicates statistically significant variation in the observed effect sizes beyond variation due to random sampling error ($Q = 695.7$, $df = 151$, $p < .001$). The percentage of variation in observed effects that is due to heterogeneity rather than sampling error is $I^2 = 83.1\%$.

Risk of bias

Methodological Quality was assessed using the Jadad Instrument.¹⁷³ Each study was evaluated for randomness, blinding, and attrition and given a final rating of 0–3 (with a high score indicating low risk of bias). The risk of bias assessment for each study is presented in table 1. The mean rating score was 1.5, indicating that on average, studies would benefit from using more rigorous methodological procedures to reduce the risk of bias.

Subgroup and meta-regression analyses

Because there is a statistically significant level of heterogeneity, subgroup and meta-regression analyses were performed on the primary studies' observed effect sizes to identify sources of the heterogeneity.¹⁸¹ Both relaxation and clinical diagnosis of depression were found to be statistically significant contributors to heterogeneity in effect sizes. The mean effect size was larger for studies that utilized relaxation than for those that did not (0.62 vs 0.34, respectively, $Q = 5.5$, $p = 0.019$). The mean effect size was larger for studies whose participants were clinically diagnosed with depression than for studies whose participants were not clinically diagnosed (0.78 vs 0.48, respectively, $Q = 6.0$, $p = 0.014$). Table 2 lists the results of the subgroup analyses for all eight qualitative moderator variables. The meta-regression indicates the effect size increases statistically significantly with the length of treatment ($\beta_1 = 0.005$, $SE = 0.001$, $z = 3.65$, $p < .001$). A scatterplot of the meta-regression results appears in Figure 4.

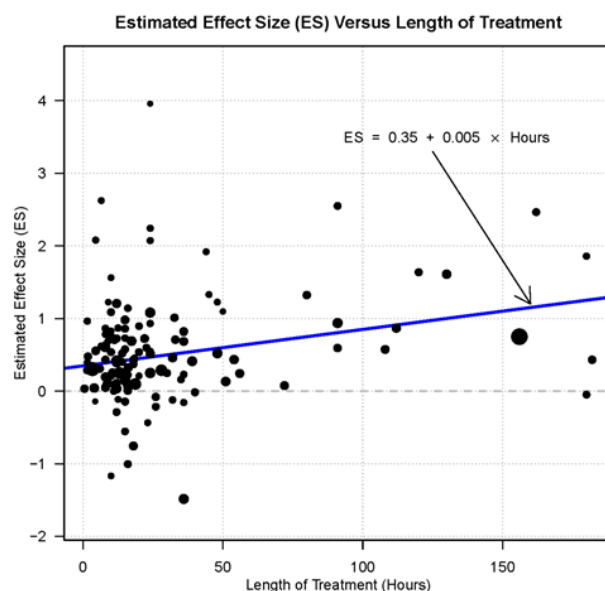


Figure 4 Scatterplot of the Meta-Regression for Effect Size versus Length of Treatment.

Publication bias assessment

The funnel plot in Figure 5 shows very little tendency for studies with larger standard errors (smaller sample sizes) to be missing due

to null findings (i.e., no evidence for publication bias). Although the trim-and-fill procedure produced an estimate of three missing studies on the left side of the funnel plot ($SE = 2.83$), the corresponding

hypothesis test found no significant evidence that the actual number of missing studies is different from zero ($p = 0.063$). The fail-safe n , using Rosenberg's method, was found to be 21782.

Table 2 Subgroup Analysis Results

Moderator Variable	n	Subgroup	Mean ES	SE	95% Confidence Interval		Wald Q Test	
					From	To	Q (df)	p-value
Type of Yoga	114	Other	0.51	0.06	0.39	0.63	12.4 (6)	0.054
	11	Swami Vivekananda Yoga	1.02	0.19	0.64	1.41		
	2	Inner Resources Program	0.49	0.47	-0.43	1.42		
	5	Kripalu	0.07	0.27	-0.46	0.60		
	17	Iyengar	0.74	0.17	0.41	1.07		
Breathing	1	Bikram Yoga	0.66	0.57	-0.46	1.77	0.35 (1)	0.551
	2	Kundalini	0.02	0.44	-0.84	0.88		
	18	No breathing	0.46	0.16	0.15	0.77		
Meditation	134	Breathing	0.56	0.06	0.45	0.67	0.08 (1)	0.780
	38	No meditation	0.52	0.11	0.31	0.73		
Relaxation	114	Meditation	0.56	0.06	0.44	0.67	5.50 (1)	0.019
	39	No relaxation	0.34	0.10	0.13	0.54		
Yoga Instructor's Expertise	113	Relaxation	0.62	0.06	0.50	0.74	7.02 (3)	0.071
	40	No certification	0.59	0.10	0.39	0.79		
	72	Hours not stated	0.58	0.08	0.43	0.73		
Diagnosis of Depression	22	200+ hours of training	0.23	0.14	-0.03	0.50	5.99 (1)	0.014
	18	500+ hours of training	0.72	0.15	0.42	1.03		
Type of Control Group	115	Not clinical	0.48	0.06	0.36	0.59	3.30 (2)	0.192
	37	Clinical	0.78	0.11	0.57	1.00		
Study Design	54	No Treatment	0.50	0.09	0.33	0.67	0.43 (1)	0.510
	61	Comparison Group	0.48	0.09	0.31	0.65		
	37	Repeated Measures	0.71	0.10	0.50	0.91		
	115	Non-RCT	0.59	0.09	0.42	0.77		
	37	RCT	0.52	0.07	0.39	0.65		

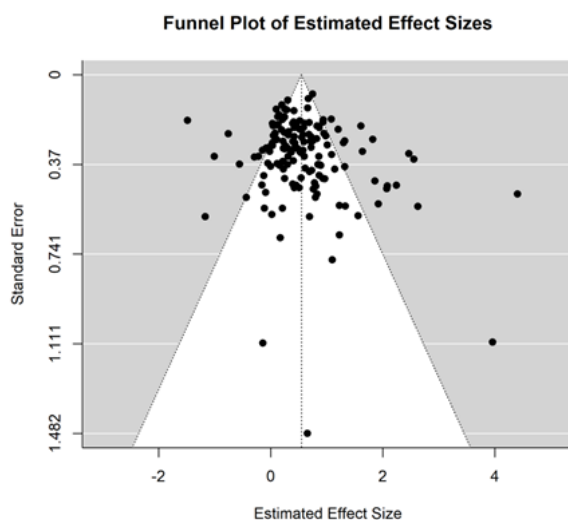


Figure 5 Funnel plot of the 152 primary studies.

Sensitivity analysis

All but nine of the studies included in the meta-analysis used a repeated measures study design (PP or PPC). To determine the sensitivity of the meta-analysis results to the assumption of a within-subject correlation value $r_1 = 0.5$, the analysis was repeated using $r_1 = 0.2$ and $r_1 = 0.8$. In both cases, the overall effect, and subgroup and meta-regression analyses results remained essentially unchanged.

Only 14 studies reported multiple different measures of depression for each participant. To determine the sensitivity of the meta-analysis results to the assumption of a within-subject correlation value of $r_2 = 0.50$, the analysis was repeated using $r_2 = 0.20$ and $r_2 = 0.80$. The results were again found to be essentially unchanged.

Discussion

This meta-analysis provides evidence for the efficacy of yoga in the treatment of depression. A mean effect size of $g = 0.55$ was found with a 95% confidence interval of 0.44–0.65. According to standards outlined by Cohen,¹⁸² this is classified as a medium effect suggesting that yoga has a moderate effect for treating depression. The relatively narrow confidence interval suggests the estimated mean effect size does not contain a high degree of variance and that the true effect is likely to be moderate.

Nine potential moderating variables were examined: type of yoga practiced, components of yoga, expertise of yoga instructor, depression diagnosis, type of comparison group, research design, and length of treatment. Three significant moderators were found. First, the effect was larger when yoga was practiced with a focus on relaxation compared to studies without a stated focus on relaxation. Though other authors have suggested that breath work is particularly helpful for reducing depression symptoms,¹⁸³ the moderator analysis in the current study did not demonstrate differential effects of breathing or meditative components of yoga practice. Second, the prediction was that yoga would be more impactful for non-clinically diagnosed participants; however, the analysis demonstrated the opposite effect. Three possible reasons that yoga was found to be more impactful for clinically depressed individuals include: another variable could be

confounded with diagnosis; regression to the mean; or the true effect is larger for individuals clinically depressed compared to individuals with milder symptoms.

The third significant moderator indicated the mean effect size was larger when higher amounts of yoga were practiced. This finding aligns with the prediction based on prior research showing a relationship between total number of hours practicing yoga and treatment benefits for depression.^{78,7} The relationship between duration of yoga training and intervention efficacy has also been found in the treatment of anxiety.¹⁸⁴ Furthermore, Harkess and colleagues¹⁸⁵ suggested that a minimum of once-a-week hour-long yoga sessions for a period of 8 weeks is necessary for treating chronic stress in nonclinical populations.

The remaining moderator variables were not significant, which could be the result of a low number of studies in some of the categories (e.g., for type of yoga, some categories only had one or two studies). It is possible, however, that these moderators do not play a role in the efficacy of yoga in treating depression. It is important to point out that there was no moderating effect based on RCTs versus non-RCT's. To the author's knowledge, this is the first meta-analysis of yoga interventions to include non-RCT's.

No evidence of publication bias, which occurs when studies with null findings are not published either due to author inaction or editor dismissal, was found. The funnel plot and trim-and-fill procedure did not show evidence of studies with small sample sizes being unpublished. Rosenberg's fail-safe number was 21782, indicating that more than twenty thousand studies with an average effect size of zero would have to be added to the analysis before the overall summary effect would become statistically nonsignificant. In other words, because it is unlikely that more than twenty thousand studies went unpublished, it is highly unlikely that the observed mean effect size is due to publication bias. Borenstein et al.¹⁵ suggested that a goal of a publication bias assessment should be to classify the bias into one of three categories: (a) where the impact of the bias is trivial; (b) where the impact is not trivial, but the major finding is still valid; and (c) where the major finding might be called into question. Based on the trim-and-fill and fail-safe n of this meta-analysis, the bias falls into category a. Therefore, the impact of publication bias appears to be trivial and does not appear to have influenced the finding that yoga can have a positive effect on depression.

Limitations

Because this meta-analysis combines studies that used varying measures, study designs, and populations, some might argue it is inappropriate to lump together such discrepant studies. As Gotzsche¹⁸⁶ noted, researchers can be categorized as "lumpers" and "splitters." The intention of this analysis was to lump together all available studies to increase the power to examine potential moderator variables. It is acknowledged that some researchers may not share this ideological approach to conducting a meta-analysis. Additionally, the results of meta-analyses are limited by the quality of the individual studies.¹⁸⁷ Therefore, the results are limited by methodological weaknesses that may exist within the various studies, such as participants who were not blind to the procedure and a failure to report withdrawals and dropouts.¹⁷³

Future research

While the current study found a larger effect for treating depression when yoga was focused on relaxation, future research

should further explore whether various aspects, or "limbs," of yoga are more impactful for improving mood. Further, more research is needed to understand the psychological and biological mechanisms of change that occur from practicing yoga and why they are related to improving depression symptoms. Several authors have hypothesized that yoga may decrease depression due to improvements in autonomic nervous system regulation.^{188,189} The current study found that yoga was more beneficial when focused on relaxation, which may be due to the increased parasympathetic response; however, this hypothesis needs to be further explored. Other underlying mechanisms that have been hypothesized include changes in brain chemistry and decreased inflammation.¹⁹⁰ Despite some progress in examining mechanisms, more research is needed to understand how yoga reduces depression symptoms.

Two additional areas that warrant further investigation are studies examining the efficacy of yoga for children and adolescent samples and long-term follow-up studies. While some studies have examined the impact of yoga on quality-of-life measures for youth,¹⁹¹ yoga should be further explored for treating a wide range of mental health challenges for children and adolescents. In addition, more research about the long-term benefits of yoga practice is needed. It is recommended that researchers report remission rates. Currently, few yoga studies were found to have reported long-term follow-up data.⁷

There is a growing use of yoga therapy which entails working with a practitioner who has received specialized training to increase a client's self-awareness and redirect energy.¹⁹² Yoga therapy involves a yoga therapist whereas yoga as an intervention involves a yoga teacher. Further, there have been recent efforts to examine the use of psychotherapy in conjunction with a yoga intervention.⁷⁷ Kenny¹⁹³ described the blending of these two therapeutic approaches. Presumably, the combination of these two therapeutic approaches might produce even greater benefits to the patient.¹⁹⁴ It would be beneficial to examine the relative efficacy of three types of yoga interventions: yoga intervention, yoga in conjunction with psychotherapy, and yoga intervention without a specialized practitioner.

Policy and practice

Beyond informing future research, this study can help to inform social, political, and organizational policy. The significant effect of yoga for treating depression means that such a treatment approach should be taken seriously. Based on the meta-analytic findings of this study, mental health professionals are encouraged to utilize yoga as an integrative or complementary approach to intervention. Along with this, insurance companies would be wise to pay for yoga interventions for depression, and universities and institutions should consider training mental health care providers in the implementation of yoga. Finally, politicians should enact legislation that fosters the availability of holistic health treatments such as yoga.

Conclusion

In sum, the meta-analysis of 152 independent studies ($n = 8210$) that examined yoga practice for reducing depression produced a moderate mean effect size ($g = .55$). The 95% confidence interval 0.44–0.65 suggests that the true mean value in the population is likely to be moderate. These findings indicate that yoga is effective for reducing depression symptoms and should be considered as an integrative treatment. Three of the nine moderator variables examined were significant. The effect was larger when yoga was focused on relaxation, for individuals with a clinical diagnosis, and when treatment occurred for longer amounts of time.

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Conflicts of interest

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